

AQA A-Level Further Mathematics (7367)

Question Paper

Continuous Random Variables

SD4 – Mean; variance + S.D.
for CRV

(A-Level)

Name: _____

Class: _____

Date: _____

Time: **410 min.**

Marks: **323 marks**

Comments:

Q1.

Participants in a school jumping competition gain a total score for each jump based on the length, L metres, jumped beyond a fixed point and a mark, S , for style.

L may be regarded as a continuous random variable with probability density function

$$f(l) = \begin{cases} wl & 0 \leq l \leq 15 \\ 0 & \text{otherwise} \end{cases}$$

where w is a constant.

S may be regarded as a discrete random variable with probability function

$$P(S = s) = \begin{cases} \frac{1}{15}s & s = 1, 2, 3, 4, 5 \\ 0 & \text{otherwise} \end{cases}$$

Assume that L and S are independent.

The total score for a participant in this competition, T , is given by $T = L^2 + \frac{1}{2}S$

Show that the expected total score for a participant is $114\frac{1}{3}$

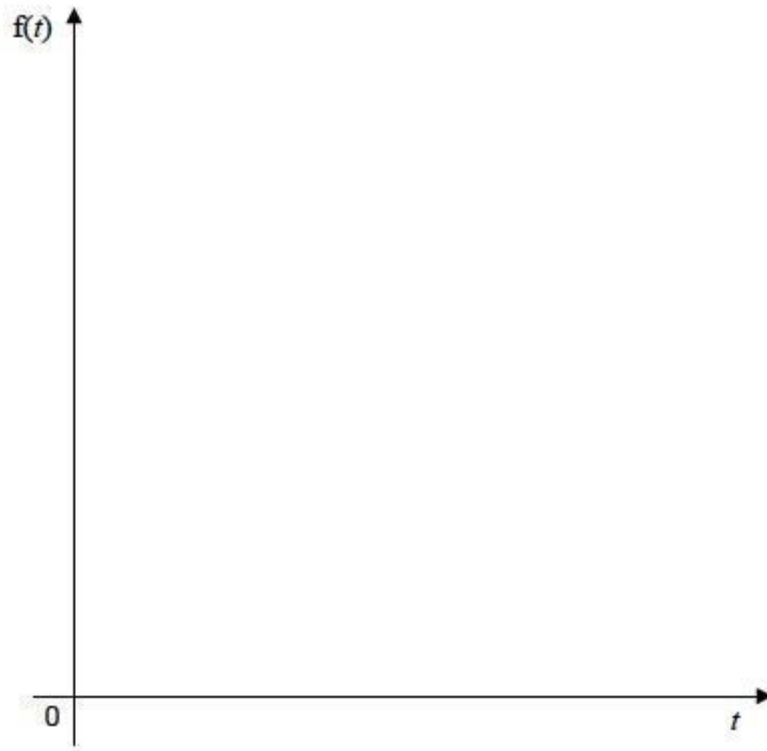
(Total 5 marks)

Q2.

The continuous random variable T has probability density function defined by

$$f(t) = \begin{cases} \frac{1}{3} & 0 \leq t \leq \frac{3}{2} \\ \frac{9-2t}{18} & \frac{3}{2} \leq t \leq \frac{9}{2} \\ 0 & \text{otherwise} \end{cases}$$

(a) (i) Sketch this probability density function below.



- (ii) State the median of T .
- (b) (i) Find $E(T)$
- (ii) Given that $E(T^2) = \frac{15}{4}$, find $\text{Var}(4T - 5)$

(2)

(1)

(2)

(3)

(Total 8 marks)

Q3.

The time, in weeks, that a patient must wait to be given an appointment in Holmsoon Hospital may be modelled by a random variable T having the cumulative distribution function

$$F(t) = \begin{cases} 0 & t < 0 \\ \frac{t^3}{216} & 0 \leq t \leq 6 \\ 1 & t > 6 \end{cases}$$

- (a) Find, to the nearest day, the time within which 90 per cent of patients will have been given an appointment.
- (b) Find the probability density function of T for all values of t .

(3)

(3)

- (c) Calculate the mean and the variance of T . (6)
- (d) Calculate the probability that the time that a patient must wait to be given an appointment is more than one standard deviation above the mean. (4)
- (Total 16 marks)

Q4.

The error, in minutes, made by Paul in estimating the time that he takes to complete a college assignment may be modelled by the random variable T with probability density function

$$f(t) = \begin{cases} \frac{1}{30} & -5 \leq t \leq 25 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find:
- (i) $E(T)$; (1)
- (ii) $\text{Var}(T)$. (1)
- (b) Calculate the probability that Paul will make an error of magnitude at least 2 minutes when estimating the time that he takes to complete a given assignment. (3)
- (Total 5 marks)

Q5.

The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} \frac{1}{2}(x^2 + 1) & 0 \leq x \leq 1 \\ (x - 2)^2 & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (3)
- (b) Calculate $P(X \leq 1)$. (3)
- (c) Show that $E(X^2) = \frac{4}{5}$. (5)

- (d) (i) Given that $E(X) = \frac{19}{24}$ and that $\text{Var}(X) = \frac{499}{k}$, find the numerical value of k . (3)
- (ii) Find $E(5X^2 + 24X - 3)$. (2)
- (iii) Find $\text{Var}(12X - 5)$. (2)
- (Total 18 marks)

Q6.

The continuous random variable X has the cumulative distribution function

$$F(x) = \begin{cases} 0 & x < -c \\ \frac{x+c}{4c} & -c \leq x \leq 3c \\ 1 & x > 3c \end{cases}$$

where c is a positive constant.

- (a) Determine $P\left(-\frac{3c}{4} < X < \frac{3c}{4}\right)$. (2)
- (b) Show that the probability density function, $f(x)$, of X is
- $$f(x) = \begin{cases} \frac{1}{4c} & -c \leq x \leq 3c \\ 0 & \text{otherwise} \end{cases}$$
- (2)
- (c) Hence, or otherwise, find expressions, in terms of c , for:
- (i) $E(X)$; (1)
- (ii) $\text{Var}(X)$. (1)
- (Total 6 marks)

Q7.

The continuous random variable X has probability density function given by

$$f(x) = \begin{cases} \frac{1}{2} & 0 \leq x \leq 1 \\ \frac{3-x}{4} & 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (3)
- (b) Explain why the value of η , the median of X , is 1. (2)
- (c) Show that the value of μ , the mean of X , is $\frac{13}{12}$. (4)
- (d) Find $P(X < 3\mu - \eta)$. (3)
- (Total 12 marks)

Q8.

The waiting time, X minutes, for fans to gain entrance to see an event may be modelled by a continuous random variable having the distribution function defined by

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2}x & 0 \leq x \leq 1 \\ \frac{1}{54}(x^3 - 12x^2 + 48x - 10) & 1 \leq x \leq 4 \\ 1 & x > 4 \end{cases}$$

- (a) (i) Sketch the graph of F . (4)
- (ii) Explain why the value of q_1 , the lower quartile of X , is $\frac{1}{2}$. (2)
- (iii) Show that the upper quartile, q_3 , satisfies $1.6 < q_3 < 1.7$. (3)
- (b) The probability density function of X is defined by

$$f(x) = \begin{cases} \alpha & 0 \leq x \leq 1 \\ \beta(x-4)^2 & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (i) Show that the **exact** values of α and β are $\frac{1}{2}$ and $\frac{1}{18}$ respectively. (5)
- (ii) Hence calculate $E(X)$. (5)
- (Total 19 marks)

Q9.

The continuous random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0 & x < -1 \\ \frac{x+1}{k+1} & -1 \leq x \leq k \\ 1 & x > k \end{cases}$$

where k is a positive constant.

- (a) Find, in terms of k , an expression for $P(X < 0)$. (2)
- (b) (i) Show that the probability density function of X is defined by
- $$f(x) = \begin{cases} \frac{1}{k+1} & -1 \leq x \leq k \\ 0 & \text{otherwise} \end{cases}$$
- (2)
- (ii) Sketch the graph of f . (2)
- (c) Given that $\text{Var}(X) = [E(X)]^2$, show that $k = 2 \pm \sqrt{3}$. (5)
- (Total 11 marks)

Q10.

The delay, in hours, of certain flights from Australia may be modelled by the continuous random variable T , with probability density function

$$f(t) = \begin{cases} \frac{2}{15}t & 0 \leq t \leq 3 \\ 1 - \frac{1}{5}t & 3 \leq t \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (3)

- (b) Calculate:
- (i) $P(T \leq 2)$; (2)
- (ii) $P(2 < T < 4)$. (3)
- (c) Determine $E(T)$. (4)
- (Total 12 marks)

Q11.

The continuous random variable X has cumulative distribution function

$$F(x) = \begin{cases} 0 & x < -1 \\ \frac{x+1}{k+1} & -1 \leq x \leq k \\ 1 & x > k \end{cases}$$

where k is a positive constant.

- (a) Find, in terms of k , an expression for $P(X < 0)$. (2)
- (b) Determine an expression, in terms of k , for the lower quartile, q_1 . (3)
- (c) Show that the probability density function of X is defined by
- $$f(x) = \begin{cases} \frac{1}{k+1} & -1 \leq x \leq k \\ 0 & \text{otherwise} \end{cases}$$
- (2)
- (d) Given that $k = 11$:
- (i) sketch the graph of f ; (2)
- (ii) determine $E(X)$ and $\text{Var}(X)$; (2)
- (iii) show that $P(q_1 < X < E(X)) = 0.25$. (2)

(Total 13 marks)

Q12.

The continuous random variable X has the cumulative distribution function

$$F(x) = \begin{cases} 0 & x \leq -4 \\ \frac{x+4}{9} & -4 \leq x \leq 5 \\ 1 & x \geq 5 \end{cases}$$

- (a) Determine the probability density function, $f(x)$, of X . (2)
- (b) Sketch the graph of f . (2)
- (c) Determine $P(X > 2)$. (2)
- (d) Evaluate the mean and variance of X . (2)
- (Total 8 marks)

Q13.

Students are each asked to measure the distance between two points to the nearest tenth of a metre.

- (a) Given that the rounding error, X metres, in these measurements has a rectangular distribution, explain why its probability density function is

$$f(x) = \begin{cases} 10 & -0.05 < x \leq 0.05 \\ 0 & \text{otherwise} \end{cases}$$

- (b) Calculate $P(-0.01 < X < 0.02)$. (2)
- (c) Find the mean and the standard deviation of X . (2)
- (Total 7 marks)

Q14.

The continuous random variable X has the probability density function given by

$$f(x) = \begin{cases} 3x^2 & 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Determine:

(i) $E\left(\frac{1}{X}\right)$;

(3)

(ii) $\text{Var}\left(\frac{1}{X}\right).$

(4)

(b) Hence, or otherwise, find the mean and the variance of $\left(\frac{5+2X}{X}\right).$

(5)

(Total 12 marks)

Q15.

Engineering work on the railway network causes an increase in the journey time of commuters travelling into work each morning.

The increase in journey time, T hours, is modelled by a continuous random variable with probability density function

$$f(t) = \begin{cases} 4t(1-t^2) & 0 \leq t \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Show that $E(T) = \frac{8}{15}.$

(3)

(b) (i) Find the cumulative distribution function, $F(t)$, for $0 \leq t \leq 1.$

(2)

(ii) Hence, or otherwise, for a commuter selected at random, find

$$P(\text{mean} < T < \text{median})$$

(5)

(Total 10 marks)

Q16.

The error, R centilitres, made in measuring the contents of bottles of wine can be modelled by the rectangular distribution

$$f(r) = \begin{cases} \alpha & -0.4 < r < 0.4 \\ 0 & \text{otherwise} \end{cases}$$

where α is a positive constant.

(a) Find the value of $\alpha.$

(1)

(b) Calculate the mean and the standard deviation of $R.$

(3)

(c) For a bottle of wine selected at random, calculate the probability that the **magnitude** of the error made in measuring its contents is less than 0.3 cl.

(2)
(Total 6 marks)

Q17.

The time, T hours, that the supporters of Bracken Football Club have to queue in order to obtain their Cup Final tickets has the following probability density function.

$$f(t) = \begin{cases} \frac{1}{5} & 0 \leq t < 3 \\ \frac{1}{45}t(6-t) & 3 \leq t \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of f . (3)
 - (b) Write down the value of $P(T = 3)$. (1)
 - (c) Find the probability that a randomly selected supporter has to queue for at least 3 hours in order to obtain tickets. (2)
 - (d) Show that the median queuing time is 2.5 hours. (2)
 - (e) Calculate $P(\text{median} < T < \text{mean})$. (6)
- (Total 14 marks)

Q18.

The continuous random variable X has the following cumulative distribution function.

$$f(x) = \begin{cases} 0 & x < 0 \\ \frac{x^3}{64} & 0 \leq x \leq 4 \\ 1 & x > 4 \end{cases}$$

- (a) Calculate $P(2 \leq X \leq 3)$. (2)
- (b) Find the probability density function, $f(x)$, for $0 \leq x \leq 4$. (2)
- (c) Evaluate:
 - (i) $E(X)$; (3)
 - (ii) $\text{Var}(X)$. (3)

Q19.

The time, T minutes, that Suneil waits to access a telephone banking service may be modelled by the following probability density function.

$$f(t) = \begin{cases} \frac{6}{19}t & 0 \leq t \leq 2 \\ \frac{6}{19}t(3-t) & 2 \leq t \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Sketch the graph of $f(t)$.

(2)

- (b) Show that the probability that Suneil waits less than 30 seconds is 0.0395, correct to three significant figures.

(2)

- (c) Calculate Suneil's mean waiting time.

(5)

(Total 9 marks)

Q20.

The time, T hours, taken by any member of a group of friends to complete a run for charity can be modelled by the following probability density function.

$$f(t) = \begin{cases} \frac{1}{90}t^2 & 3 \leq t \leq 6 \\ 2 - \frac{4}{15}t & 6 \leq t \leq 7.5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) (i) Show that the probability that a member of the group selected at random takes at least 6 hours to complete the run is 0.3.

(2)

- (ii) Evaluate the median time taken to complete the run.

(6)

- (b) Calculate the mean time taken to complete the run.

(5)

(Total 13 marks)

Q21.

At a Post Office, the time, T minutes, that customers have to wait in order to be served has the following probability density function.

$$f(t) = \begin{cases} \frac{t^2}{18} & 0 \leq t < 3 \\ \frac{1}{4}(5-t) & 3 \leq t \leq 5 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Write down the value of $P(T = 4)$. (1)
- (b) Show that the median waiting time is 3 minutes. (2)
- (c) Find the probability that customers have to wait for less than 3.5 minutes to be served. (5)
- (d) Calculate the mean time that customers have to wait in order to be served. (5)
- (Total 13 marks)

Q22.

The delay, T minutes, in connecting electronically to a ticket booking service can be modelled by the distribution function:

$$P(T \leq t) = 1 - e^{-2t}, \quad t \geq 0.$$

- (a) (i) Write down the probability density function of T . (1)
- (ii) State values for the mean and standard deviation of T . (2)
- (b) Calculate the probability that the delay exceeds 3 minutes. (2)
- (c) Find the median delay, giving your answer in seconds, to the nearest second. (3)
- (Total 8 marks)

Q23.

The continuous random variable X has a rectangular distribution over the interval a to $a + k$, where a and k are positive constants.

- (a) Find, in terms of a and k , expressions for the mean and variance of X . (2)
- (b) Given that $E(X) = 21$ and $\text{Var}(X) = 3$, show that $k = 6$ and $a = 18$. (3)

- (c) Hence calculate $P(20 < X < 25)$

(3)

(Total 8 marks)

Q24.

The continuous uniform random variable X has the following probability density function.

$$f(x) = \begin{cases} \frac{1}{6} & 6 \leq x \leq 12 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find values for the mean, μ , and variance, σ^2 , of X .

(2)

- (b) Determine $P(X > \mu + \sigma)$.

(3)

(Total 5 marks)

Q25.

The continuous random variable X has the following probability density function.

$$f(x) = \begin{cases} kx & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that the value of the constant k is $\frac{1}{8}$.

(2)

- (b) Given that $E(X) = 2\frac{2}{3}$, calculate the variance of X .

(4)

- (c) Hence calculate $\text{Var}(5X - 4)$.

(2)

- (d) Calculate the 90th percentile of X .

(3)

(Total 11 marks)

Q26.

The lengths, X centimetres, of boards used in garden fencing are uniformly distributed over the interval 198 to 204.

- (a) Find values for the mean, μ , and the variance, σ^2 , of X .

(2)

- (b) Write down the probability density function, $f(x)$, of X for $198 \leq x \leq 204$.

(1)

(c) Determine:

(i) $P(X = \mu)$;

(1)

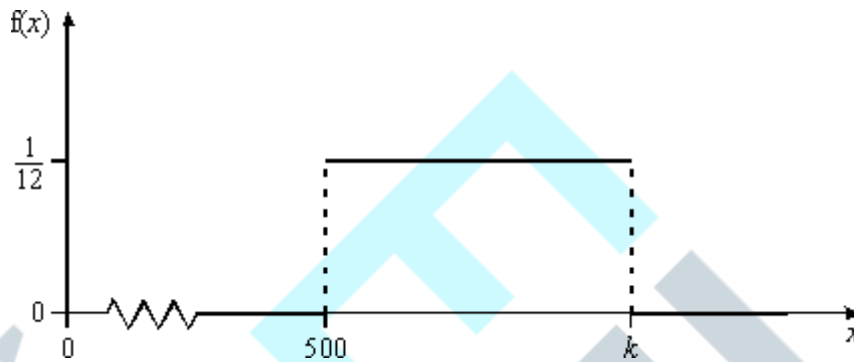
(ii) $P(X < 200 - \sigma)$.

(3)

(Total 7 marks)

Q27.

The probability density function, $f(x)$, for the weight, X grams, of marshmallow pieces in a tub is shown by the following graph.



(a) State why the value of k on the horizontal axis is 512.

(1)

(b) Determine values for the mean, μ , and standard deviation, σ , of X .

(3)

(c) Calculate:

(i) $P(X < 509)$;

(2)

(ii) $P(X > \mu - \sigma)$.

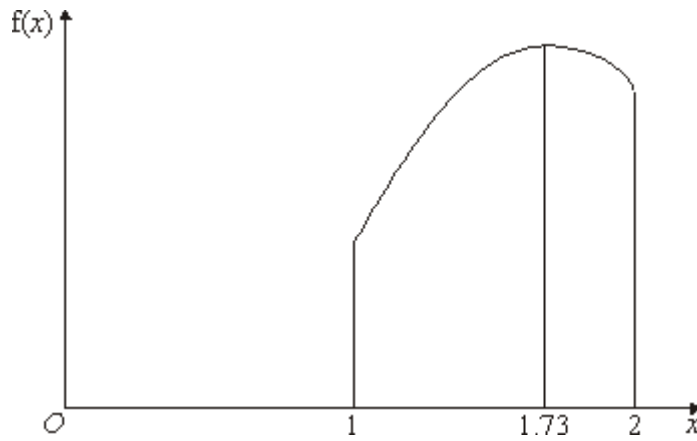
(3)

(Total 9 marks)

Q28.

The sketch below shows the probability density function $f(x)$ of the continuous random variable X .

($f(x) = 0$ outside the interval $1 \leq x \leq 2$.)



You are given that $\int_1^{1.58} f(x) dx = 0.5$ $\int_{1.73}^2 f(x) dx = 0.291$

$$\int_1^2 xf(x) dx = 1.54 \quad \int_1^2 (x - 1.54)^2 f(x) dx = 0.0751$$

(a) Write down the value of:

(i) $\int_1^2 f(x) dx$;

(ii) the mean of X ;

(iii) the median of X ;

(iv) the mode of X ;

(v) the range of X .

(5)

(b) Find:

(i) the probability that X lies between the median and the mode;

(2)

(ii) the standard deviation of X ;

(2)

(iii) $E(X^2)$.

(2)

(Total 11 marks)

Q29.

The random variable X has probability density function:

$$f(x) = \begin{cases} px+q & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

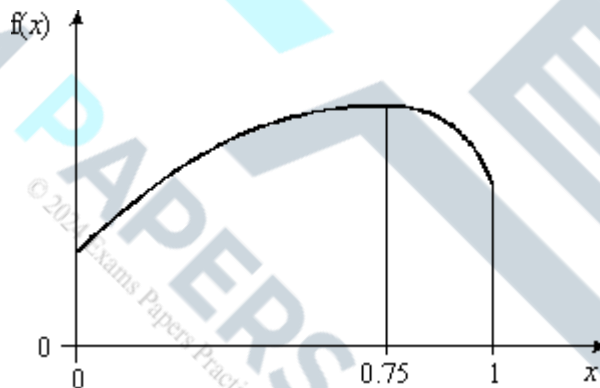
where p and q are constants.

- (a) Show that $\frac{1}{2}p + q = 1$. (3)
- (b) Find the mean of X in terms of p and q . (3)
- (c) The mean of X is 0.6. Show that $p = 1.2$ and find the value of q . (3)
- (d) Find the standard deviation of X . (4)
- (Total 13 marks)

Q30.

The sketch below shows the probability density function $f(x)$ of the continuous random variable X .

$[f(x) = 0 \text{ outside the interval } 0 \leq x \leq 1]$



- (a) Explain why the value of the mode of X is 0.75. (1)
- (b) You are given that:
- $E(X) = 0.542$ $E(X^2) = 0.369$
- $\int_0^{0.557} f(x)dx = 0.500$ $\int_0^{0.542} f(x)dx = 0.483$ $\int_0^{0.75} f(x)dx = 0.719$
- (i) Find the standard deviation of X . (3)
- (ii) Find the probability that X exceeds the mode. (2)
- (iii) Write down the value of the median of X . (1)

- (iv) Evaluate the probability that X lies between the median and the mean.

(3)

(Total 10 marks)

Q31.

The errors, in grams, made by a butcher's scales may be modelled by a random variable, y , with probability density function

$$f(y) = \begin{cases} k & -6 < y < 14 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that $k = 0.05$.

(2)

- (b) Write down the mean value of Y .

(1)

- (c) (i) Use integration to find $E(Y^2)$.

(3)

- (ii) Hence, or otherwise, find the standard deviation of Y .

(3)

- (d) (i) Find the probability that an error is negative.

(1)

- (ii) Show that the probability of an error having magnitude less than 2 is 0.2.

(2)

- (iii) Find the probability that the magnitude of an error is greater than 4.

(2)

(Total 14 marks)